

A reliable and compact STT-MRAM based digital Look-Up Table CiM

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This work proposes a novel STT-MRAM based digital Look-Up Table (LUT) Computing-in-Memory (CiM) architecture for neural networks that addresses the read and write challenges associated with high-density LUT integration. The proposed CiM technique introduces key enhancements: (1) a scalable STT-MRAM based digital Look-up table CiM architecture (2) a reliable and area-efficient readout scheme by using inverters. It is shown that the novel CiM achieves a simplified structure, a 1 ns read delay, and an average computing energy of 39.6 fJ/bit. Compared with prior state-of-the-art (SOTA) STT-MRAM-based digital CiM (DCiM) designs, the proposed architecture offers improved structure, computing speed and per bit read energy.

Index Terms—digital Computing-in-memory (DCiM), look-up table (LUT), spin-transfer torque magnetic random-access memory (STT-MRAM), high-density and high-reliability

I. INTRODUCTION

Look-up table (LUT)-based computing-in-memory (CiM) has emerged as an efficient digital CiM approach by reducing logic overhead, replacing portions of multiply-accumulate (MAC) operations with memory-based look-up operations [1], [2]. Memory density remains a key factor for scalability, motivating architectures that support a large number of compact LUT instances [2].

Spin-transfer torque magnetic random-access memory (STT-MRAM) is a promising candidate for LUT-CiM due to its nonvolatile storage and ability to store precomputed partial sums. However, achieving high-density STT-MRAM-based LUT-CiM is challenging, as efficient read and write operations must be maintained within a compact architecture.

This work proposes a dense STT-MRAM-based LUT-CiM architecture that addresses these challenges. Circuit-level simulations demonstrate scalable and efficient operation, highlighting the viability of STT-MRAM for future digital CiM systems.

II. PROPOSED STT-MRAM BASED LUT-CiM ARCHITECTURE AND ITS WORKING PRINCIPLE

The proposed LUT-CiM architecture, illustrated in Fig. 1 (a), consists of a conventional 1T-1MTJ memory array with integrated readout and logic circuits. To achieve high density, inverter-based readout is adopted. The access transistor in each bitcell is sized at three times the minimum to support the high current requirement of the STT-MRAM.

Fig. 1 (b) shows the working principle of computing. During computation, input activations are applied to the word-line (WL) decoder to select a row. The bit-line input (BLI) node is then charged, and its voltage determined by the stored MTJ state, is sensed by inverters to generate full-swing outputs (“0” or “1”). However, inverter-based sensing is challenged by the limited MTJ tunnel magnetoresistance (TMR). To mitigate this, a bias transistor is inserted on each bit line (BL). With a read voltage applied from the source line (SL), the bias transistor and bitcell form a voltage divider. Adjusting the bias voltage V_b

tunes the effective resistance, enabling reliable modulation of the BLI voltage.

During the write operation, precomputed data are stored in the MTJ using a conventional scheme: a current from SL to BL writes a “0,” while a current from BL to SL writes a “1”. Meanwhile, V_b is grounded to turn off the bias transistor. However, due to the IR drop across the MTJ, the gate-source voltage of the access transistor is lower than V_{DD} , resulting in insufficient driveability to fully turn on the pMOS transistor. Therefore, a boosted write voltage in SL and BL is required to provide a gate-source voltage exceeding V_{DD} to ensure proper operation. The boosted voltage is carefully constrained by the CMOS reliability limit of 1 V, as reported in prior work [3].

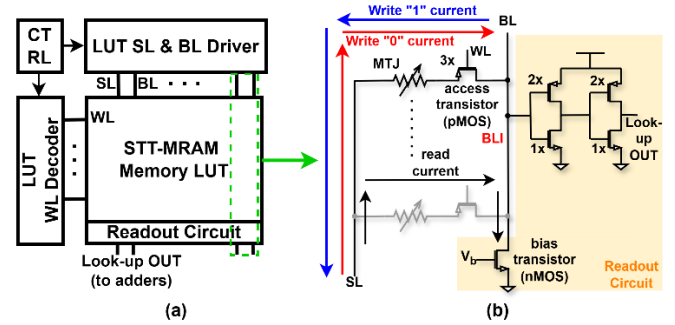


Fig. 1. (a) The architecture of proposed STT-MRAM based LUT-CiM. (b) One memory column with readout circuit of LUT-CiM.

III. EVALUATION AND BENCHMARKING

The proposed CiM cell was designed using commercial 22 nm CMOS technology. A calibrated IMEC MTJ model [3] characterized by a nominal $R_{ap} = 6.5$ k Ω and $R_p = 2.5$ k Ω is used. All simulations were conducted using the Cadence Virtuoso Spectre simulator.

Fig. 2 illustrates the read operation results when MTJ in the antiparallel (AP) and parallel (P) states, respectively. SL is set to the nominal $V_{DD} = 0.8$ V during read operation. The WL pulse width is set to 1 ns to limit read current and reduce read disturbance. V_M represents the switching voltage threshold of the output inverter. During standby, the bias voltage V_b is fixed to V_{DD} to turn on the bias transistor and reset BLI node’s voltage

value. During read operation, it is reduced to 0.7 V to change the resistance of the bias transistor, further adjusting the ratio of the voltage divider.

According to prior work [2], the optimal LUT size ranges from 4 to 16 rows. To evaluate the impact of parasitic effects, simulations were performed using a π -type resistance-capacitance (RC) network. The results show that the performance difference between 4-row and 16-row LUTs is negligible. In both cases, the sensing margins for logic “0” and “1” are approximately 130 mV, which is sufficient for reliable readout. This indicates that the proposed circuit maintains stable sensing performance across the evaluated LUT sizes and exhibits strong robustness against parasitic effects.

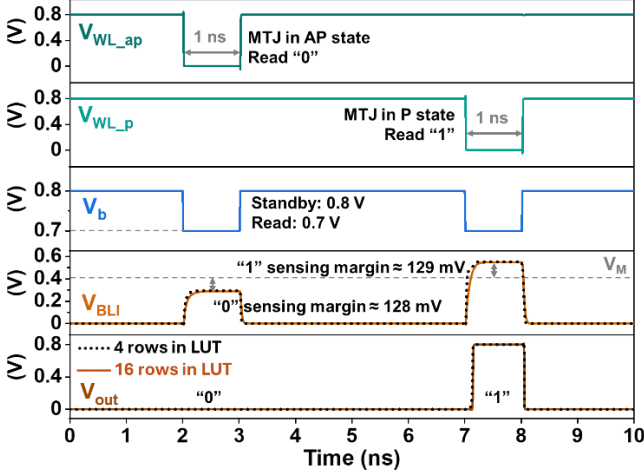


Fig. 2. Read operation waveforms of the proposed LUT-CiM.

Fig. 3(a) illustrates the read operation under different CMOS process corners. In extreme cases, such as fast P/slow N (FS) and slow P/fast N (SF), the BLI node voltage deviates significantly, leading to potential read failures. However, reliable readout can be restored by adjusting the bias voltage V_b to reconfigure the voltage divider ratio. As shown in Fig. 3(b), successful sensing is achieved by setting V_b to 0.6 V for FS and 0.8 V for SF. The corresponding operating conditions are summarized in Table I. The MTJ corners TH, TT, and TL denote typical switching speed with high, typical, and low resistance, respectively. These results demonstrate strong robustness against temperature and MTJ process variations, while providing an effective mitigation strategy for CMOS process variations.

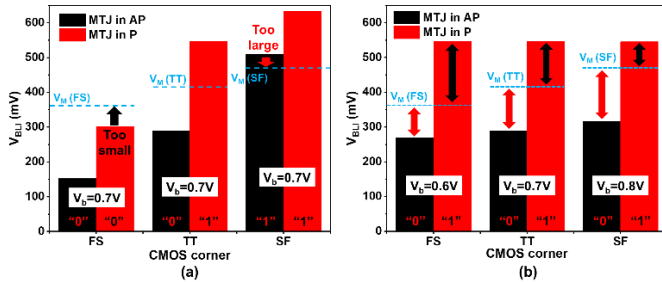


Fig. 3. (a) CMOS process corner variations' influence on read operation. (b) Results after adjusting the bias voltage.

TABLE I
WORKABLE READ CONDITIONS

Conditions	Appearance
Temperature	-40 ~ 80 °C ($V_b = 0.7$ V)
MTJ corner	TL, TT, TH ($V_b = 0.7$ V)
CMOS corner	SF, TT, FS
V_b at different CMOS corner	0.8V, 0.7V, 0.6V

Table II compared the performance of this work with some previous STT-MRAM based DCiM works. According to the table, this work shows a simpler cell and readout circuit, which will save area, and lower computing energy.

TABLE II
BENCHMARK

	This work	TCAS-II'23 [4]	TMAG'26 [5]
Process	22nm	28nm	55nm
Cell and Readout structure	1T1M + 2 inverters	1T1M + 2-stage SA	4T2M + 2T
Delay for readout (ns)	1	10 ^a	0.68
Read energy per bit (fJ/b)	39.6 ^b	163.7	109.9

^awith 1024 cells post-layout extraction. ^baverage power of read “0” and “1”.

IV. CONCLUSIONS

A novel STT-MRAM-based digital LUT-CiM architecture is proposed. SPICE simulations demonstrate that the designed CiM achieves a reliable sensing margin exceeding 130 mV, along with enhanced robustness to temperature and process variations. In addition, it features a simplified structure, a 1 ns read delay, and an average computing energy of 39.6 fJ/bit. Compared with prior SOTA STT-MRAM-based DCiM designs, the proposed architecture offers improved structure, computing speed and per bit read energy. These advantages highlight its suitability for energy- and area-efficient, high-reliability accelerator applications.

V. ACKNOWLEDGES

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REFERENCES

- [1] C. -F. Lee *et al.*, "A 12nm 121-TOPS/W 41.6-TOPS/mm² All Digital Full Precision SRAM-based Compute-in-Memory with Configurable Bit-width For AI Edge Applications," *2022 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits)*, Honolulu, HI, USA, 2022, pp. 24-25.
- [2] Y. He *et al.*, "An Area-Efficient Lookup-Table -Based eDRAM Digital CIM Macro for Neural Network Inference," *IEEE Journal of Solid-State Circuits*, vol. 61, no. 4, pp. 1697-1709, April 2026. 05. 11.
- [3] F. Garcia-Redondo *et al.*, "A Novel DTMO-driven 1T1R Bitcell for sub-10ns STT-MRAM LLC Macros at N12 Node," *2024 IEEE European Solid-State Electronics Research Conference (ESSERC)*, Bruges, Belgium, 2024, pp. 665-668.
- [4] T. Na, "Ternary Output Binary Neural Network With Zero-Skipping for MRAM-Based Digital In-Memory Computing," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 70, no. 7, pp. 2655-2659, July 2023.
- [5] Y. Wang, T. Li and T. Endoh, "A High-Reliable and High-Speed All-Digital STT-MRAM-Based Computing-in-Memory for Binary Neural Networks," *IEEE Transactions on Magnetics*, vol. 62, no. 3, pp. 1-5, March 2026, Art no. 3400505.